Reproducibility Benchmark for the MARISMA-IoT Smart Home Case Study

Detailed benchmark of assessment effort, workflow phases, quantitative outcomes, and reproducibility requirements for the laboratory Smart Home evaluation.

1. Benchmark scope and experimental basis

The benchmark compares two executions of the same risk-analysis activities. S0 is a conventional analyst-driven baseline using the same standards and references (ENISA, ISO, and NIST), but without MARISMA-IoT pre-linked catalogs and matrices and without automated recomputation in eMARISMA. S1 is the MARISMA-IoT-assisted assessment using the embedded taxonomies, relationship matrices, guided workflow, and automated computation pipeline.

Elapsed calendar duration is not used. The benchmark is based on effective person-hours. One equivalent working month (EWM) is defined as 160 person-hours, excluding waiting periods, meetings, personnel unavailability, and unrelated internal validation delays.

Reproducibility boundary. Total effective effort is available for S0 and S1, but observed person-hours are not available separately for each phase. Phase-level timing values are therefore not inferred or fabricated. This benchmark preserves the aggregate totals and defines the records required for a future phase-level replication.

Table 1. Benchmark Conditions

Element	S0 - Conventional Baseline	S1 - MARISMA-IoT
Standards / references	Same ENISA / ISO / NIST basis used in the case study.	Same standards and references.
Asset identification	Manual identification and classification.	IoT asset taxonomy and structured classification model.
Threat / control mapping	Manual documentation and spreadsheet-based cross-referencing.	Pre-linked catalogs and relationship matrices.
Risk computation	Analyst-driven calculation and cross-referencing.	Automated computation and update pipeline in eMARISMA.
Reporting	Manual consolidation of assessment outputs.	Guided workflow, dashboards, visualization, and automated reporting.
Measured effort	3.0 EWM = 480 person-hours	2.0 EWM = 320 person-hours

2. Phase-by-phase reproducibility benchmark

Both benchmark conditions use the same activity set. Table 2 decomposes that common workflow into reproducible phases and identifies the mechanism introduced by MARISMA-IoT in each phase.

Table 2. Phase-by-Phase Benchmark Protocol and Observed Effects

Phase	Activity	S0 Procedure	S1 Procedure	Observed Effect	Timing Evidence
P1	Asset identification and classification	Manual inventory, identification, and classification.	IoT-specific asset taxonomy plus structured classification model.	25% more relevant assets identified than the conventional baseline.	Aggregate only; phase hours unavailable.
P2	Identification and mapping of applicable threats	Manual threat selection and asset-threat cross-referencing.	Predefined IoT threat catalog and pre-linked Asset-Threat-Dimensions relationships.	Less manual cross-referencing and more consistent threat applicability.	Aggregate only; phase hours unavailable.
P3	Risk analysis and likelihood / impact parameterization	Analyst parameterization and manual relationship consolidation.	Predefined relationships plus automated risk computation; expert calibration remains possible.	R = P x I and automatic recalculation when inputs change.	Aggregate only; phase hours unavailable.
P4	Identification and mapping of controls	Manual control and objective / threat mapping.	Domain-Objective-Threat matrix and embedded control relationships.	Reduced mapping effort and consistent traceability.	Aggregate only; phase hours unavailable.
P5	Control selection and prioritization	Analyst-driven prioritization from manually consolidated information.	Coverage and residual-risk information support control and objective ranking.	28% improvement in control-prioritization accuracy.	Aggregate only; phase hours unavailable.
P6	Preparation of final assessment results	Manual consolidation of outputs and reporting.	Automated dashboards, heatmaps, risk summaries, and reporting.	Faster interpretation and reproducible inherent/residual comparison.	Aggregate only; phase hours unavailable.

3. Timing benchmark and direct calculations

The aggregate timing result is directly derivable because both the normalization constant and the EWM values are specified.

Table 3. Assessment-Effort Benchmark

Metric	S0	S1	Difference	Result
Equivalent working effort	3.0 EWM	2.0 EWM	-1.0 EWM	-33.3%
Effective person-hours	480 h	320 h	-160 h	-33.3%

The normalized effort for evaluation condition s is:

$$E^{EWM}_{s} = (\Sigma^{n_s}_{i=1} h_{i,s}) / H_{EWM}, \quad H_{EWM} = 160 \text{ person-hours.} \quad (1)$$

The relative effort reduction is:

$$\Delta T = (E^{EWM}_{0} - E^{EWM}_{1}) / E^{EWM}_{0} \times 100 = (3.0 - 2.0) / 3.0 \times 100 = 33.3\%. \quad (2)$$

The equivalent person-hour calculation is (480 - 320)/480 x 100 = 33.3%, corresponding to 160 effective person-hours saved in the evaluated Smart Home case.

4. Case-study outcomes

Table 4. Case-Study Outcomes and Reproducibility Status

Outcome	Comparison	Result	Reproducibility Status
Normalized assessment effort	3.0 EWM (S0) vs 2.0 EWM (S1)	33.3% reduction	Fully recomputable using 160 person-hours/EWM.
Control-prioritization accuracy	MARISMA-IoT vs independent expert reference ranking	28% improvement	Acc@k is defined, but k and the absolute S0/S1 Acc@k values are unavailable.
Mitigated risk	Inherent vs residual risk after safeguards	Up to 40% increase	Exact independent recomputation requires the underlying threat-asset risk data.
Relevant asset identification	MARISMA-IoT taxonomy vs conventional baseline	25% more relevant assets	The case-study inventory contains 32 listed assets; the absolute baseline inventory count is unavailable.
High-risk distribution	Pre-control vs post-control risk distribution	Approx. 51% average decrease in high risks	Separate distribution-level observation; it should not be conflated with the up-to-40% mitigated-risk metric.
Example analytical output	Raspberry Pi risk distribution	High = 225; Medium = 90; Low = 15	Example from the eMARISMA analytical view.

5. Reproducibility equations and decision rules

Table 5. Metrics, Equations, and Replication Rules

Metric	Equation	Parameters	Replication Rule
EWM	E_s^EWM = Σ h_i,s / H_EWM	H_EWM = 160 person-hours.	Use effective working hours only.
Effort reduction	ΔT = (E0 - E1) / E0 x 100	S0 = baseline; S1 = MARISMA-IoT.	Apply to the same set of assessment activities in both conditions.
Risk magnitude	R = P x I	P [0, 1]; I is dimension degradation.	Keep threat-asset pairs and dimension definitions consistent when comparing inherent and residual risk.
Control prioritization	Acc@k = |C_k(MARISMA-IoT) ∩ C_k(REF)| / k	C_k(REF) is the independent expert top-k reference set.	Reference evaluators must be independent from pattern-construction experts.

6. Detailed replication record for phase-level reproducibility

The following fields constitute a replication protocol, not additional case-study results.

Table 6. Minimum Evidence to Retain for a Phase-Level Replication

Record	Minimum Information to Retain
Participant / condition	Anonymous participant ID; S0 or S1; evaluator role; confirmation of independence where applicable.
P1 - Assets	Effective minutes; identified assets; classified assets; reclassifications; unmapped assets.
P2 - Threats	Effective minutes; applicable threats selected; asset-threat links created; manual corrections required.
P3 - Risk	Effective minutes; P and I values; threat-asset pairs; affected dimensions; manual overrides and rationale.
P4 - Controls	Effective minutes; mapped controls; Domain-Objective-Threat links; control-coverage values; mapping corrections.
P5 - Prioritization	Effective minutes; generated top-k ranking; independent expert top-k reference; Acc@k; selected controls.
P6 - Reporting	Effective minutes; report-generation steps; exported tables; dashboards / heatmaps used; manual post-processing.
Risk outputs	Inherent risk, residual risk, ΔR, and low / medium / high distributions by asset and by threat.
Asset coverage	Total identified assets under S0 and S1, plus common, missing, and additional assets.
Auditability	Version of the MARISMA-IoT pattern, catalogs and matrices; eMARISMA configuration; snapshot date; all parameter changes.

7. Case-study configuration required for replication

Table 7. Minimum Smart Home Configuration to Preserve Experimental Equivalence

Configuration Item	Case-Study Condition
Deployment	Functional Smart Home physically deployed in a laboratory as a controlled testbed.
Wi-Fi layer	IP-enabled IoT devices connected through a domestic wireless router using WPA2.
ZigBee layer	Low-power sensors and actuators interconnected through a ZigBee personal area network and coordinator.
Local control	Raspberry Pi 4 hosting Home Assistant; ConBee II provides ZigBee integration.
Messaging	MQTT publish/subscribe interactions used by sensors, actuators, automation logic, and monitoring components.
Application protocols	HTTP/HTTPS used for local or remote APIs, commands, telemetry, and configuration.
Cloud dependencies	Proprietary vendor clouds used for remote access, updates, registration, telemetry, mobile applications, or assistant interoperability.
Trust boundaries	Local Wi-Fi, ZigBee, MQTT, Home Assistant, and third-party cloud services form distinct but interconnected trust zones.

8. Result evidence required for full independent recomputation

Table 8. Evidence Required to Reproduce the Case-Study Quantitative Improvements Exactly

Case-Study Result	Evidence Needed for Exact Recomputation
33.3% effort reduction	Per-participant effective hours or phase logs whose totals reproduce 480 h (S0) and 320 h (S1).
28% prioritization improvement	Exact k, S0 top-k, MARISMA-IoT top-k, independent expert reference top-k, and resulting Acc@k values.
Up to 40% mitigated risk	Threat-asset pairs, P and I values, applied controls, inherent risk, residual risk, and aggregation rule.
25% asset-identification improvement	Absolute S0 and S1 inventory counts plus the list of additional assets identified under MARISMA-IoT.
Approx. 51% high-risk decrease	Pre-control and post-control distributions used to classify risk as low, medium, and high.

9. Interpretation

The directly reproducible timing result is the reduction from 480 to 320 effective person-hours, equivalent to a 33.3% decrease in normalized assessment effort. The reduction is attributed to the IoT-specific taxonomy, pre-linked catalogs and matrices, guided workflow, automated cross-referencing, computation, update mechanisms, and reporting support provided by MARISMA-IoT and eMARISMA.

The Smart Home case study also yields a 28% improvement in control prioritization, up to 40% increase in mitigated risk, 25% more relevant assets identified, and approximately 51% fewer high-risk classifications. Exact independent recomputation requires raw phase logs, expert top-k rankings, threat-asset P/I values, and inherent/residual risk tables.